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Course Code: CSA0408

1) a) Justification:

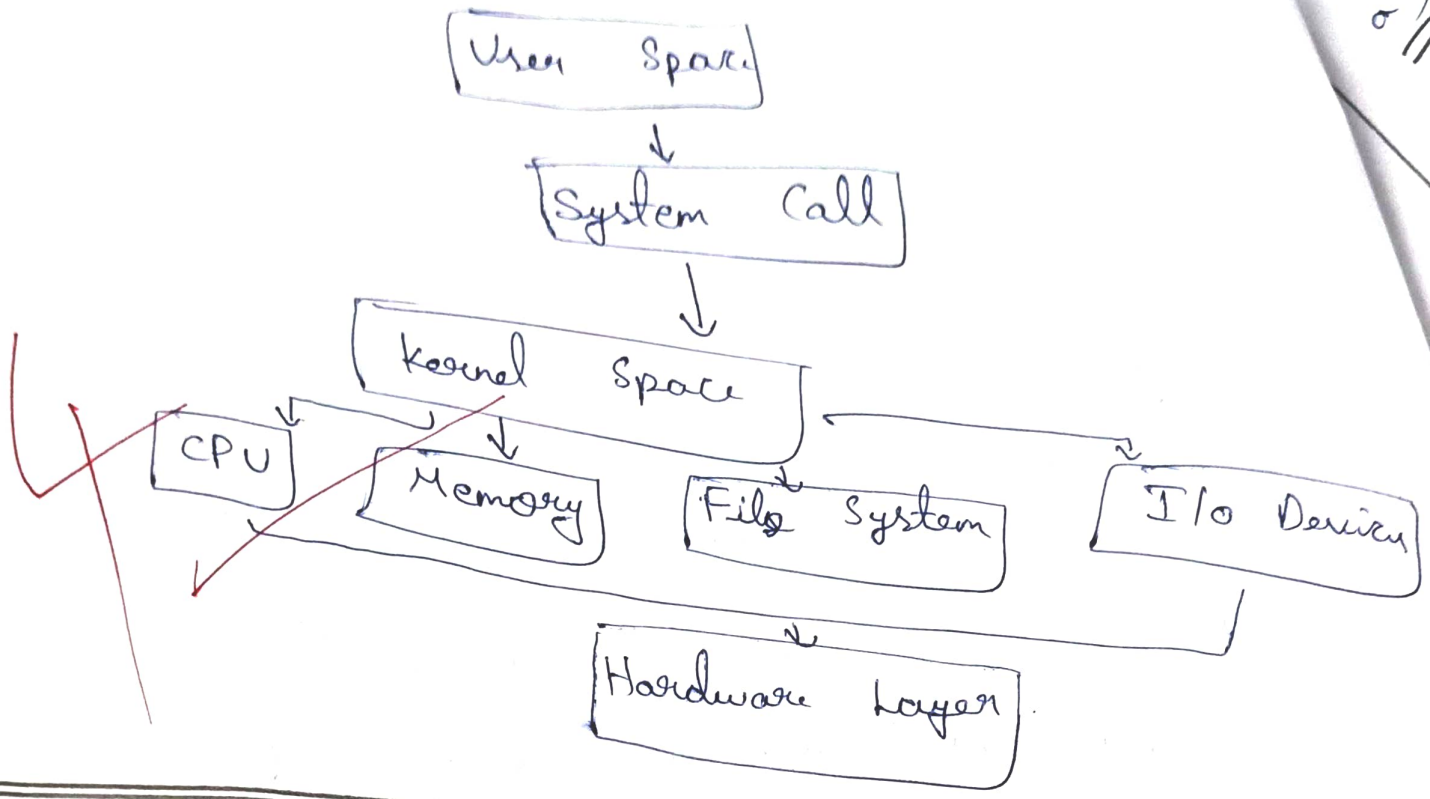
Uses high-speed CPU scheduling algorithms to switch rapidly between processes. This provides each user with a fair slice of CPU time, creating the illusion that they have executed.

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b) The layered architecture diagram below illustrates how an OS sits b/w the user space applications and the raw hardware components to safely manage an

c) 1. fork () → initiates process
2. exec () → Handles execution
3. exit () → Handles termination

a)



b)

a) File Management – Manipulates files stored on storage
Ex: Open(), read().
Device Management – Requests, controls & safely
Ex: request(), release().

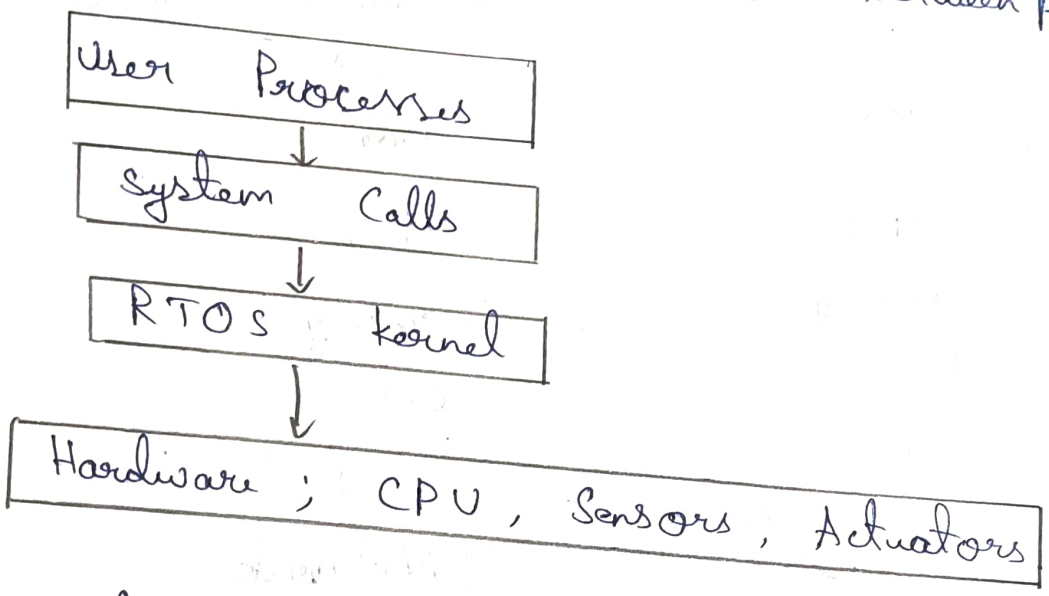
b)

Api program
↓
Logical File system
↓
File Organization Module
↓
Basic File System
↓
I/O Control
↓
Physical Device

a) A Real time OS is suitable for controlling industrial machinery

* provides fast & predictables responses
* Meets strict timing deadlines
* Supports priority based process scheduling.
* Offer regime synchronization between process

b)



c) * Sem-wait () - Waites for semaphore resource

* Sem-post () - Signals a semaphore and release access

* Pathread-mutex : lock () - Locks a shared resource.

* Pthread-mutex-unlock () - Unlocks the resource

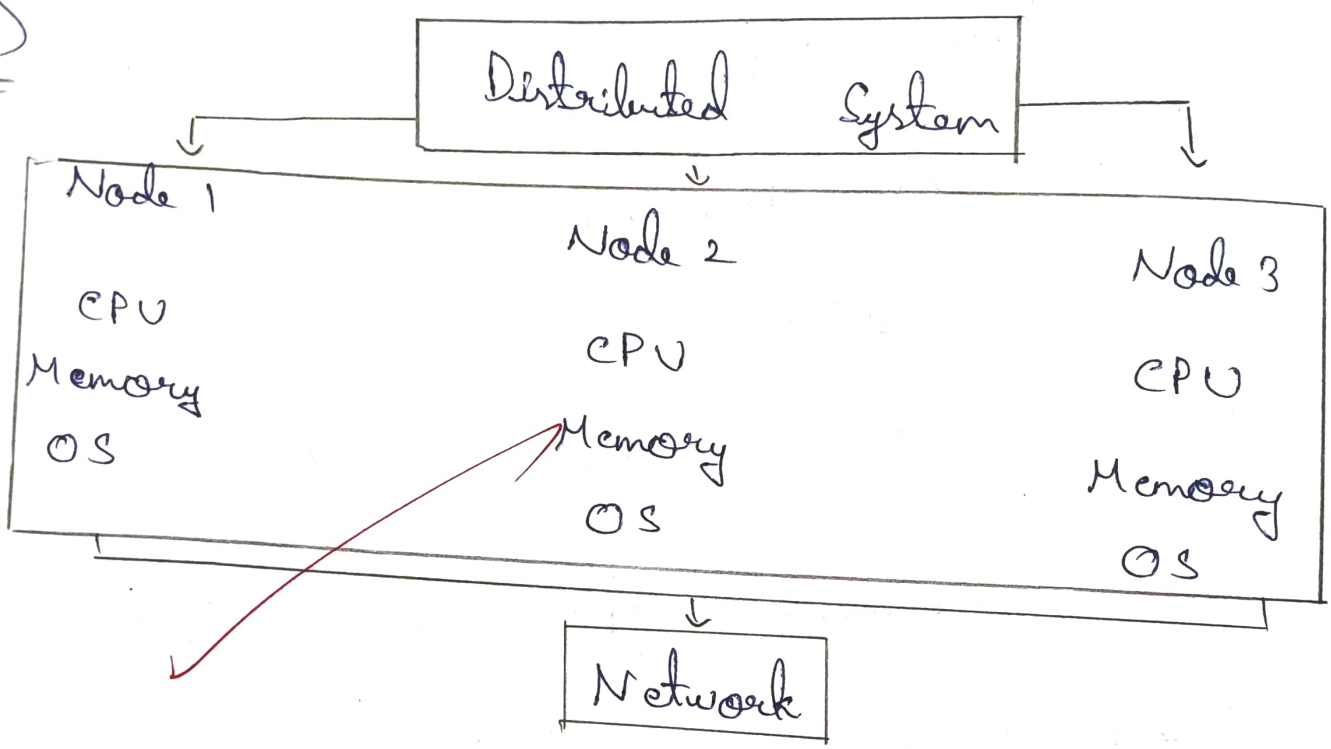
* Sched-yield () - Allows another ready process to run.

4)

a) Distributed OS

* Runs across multiple computers	* Mainly runs on one central computer
* Resource distributed	* Resource are centrally
* Provides resource sharing	* Resource are managed centr

b)



c) System Calls involved in inter-process communicat

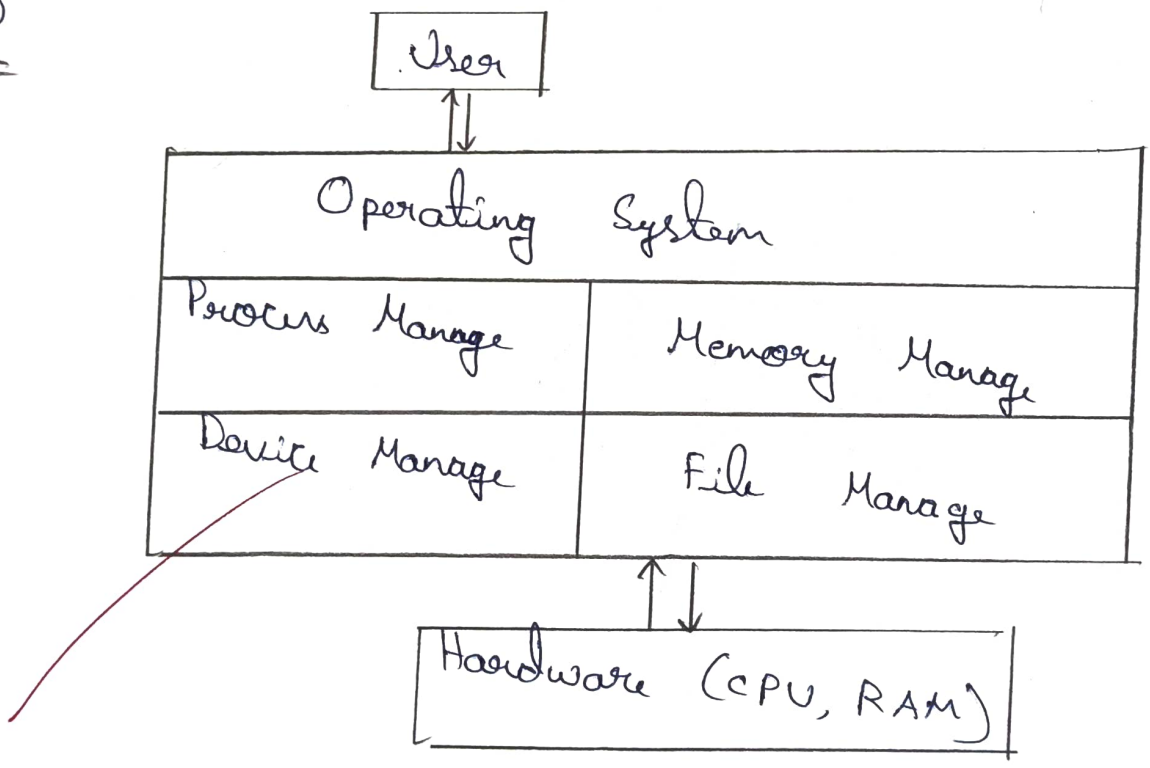
* Pipe ()
* shmget ()
* Msgsnd ()
* Socket ()
* Send ()

OS

a)

* Process Management: Manages running program & allocates CPU time
* Memory Management: Allocates & frees memory for program
* Device Management: Controls devices like printer, keyboards & disks.
* File Management: Creates, stores & delete files
* Security: Protect data from unauthorized access.

b)



c) System Call

1. Memory Allocation:

* mmap() - Allocates memory.
* munmap() - Releases allocated memory

2. Device Management:

* Open() - Opens a device.
* read() - Reads data from a device
* write() - Sends data to a device.